

Efficiency Consequences of Visit Schedule Choice in
Longitudinal Trials: An Analytic and Simulation-Based
Comparison

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Abstract

Observation times in longitudinal clinical trials are usually spaced equally for administrative convenience, but optimal design theory shows that placement affects the precision of estimated treatment effects. We compare an equally spaced five-visit schedule with a boundary-clustered schedule for a two-arm, 24-month trial with a linearly divergent treatment effect, combining an exact generalized least squares (GLS) efficiency calculation with a common-random-number Monte Carlo study (2000 paired replications) under a random-intercept, random-slope data generating model. The analytic relative efficiency of the clustered design is 1.06, and the paired simulation, after recalibrating the effect size so power is informative rather than at ceiling, reproduces this result (empirical relative efficiency 1.07, 95% bootstrap interval 1.06-1.09), confirming that the precision advantage is real and not simulation noise. We do not claim optimality: the theoretically optimal design for this covariance structure is not computed here, and only two designs are compared. The contribution is a worked, fully reproducible demonstration of how much efficiency is at stake for one realistic configuration, and a template (`R/simulation.R`) for extending the comparison to other covariance structures, dropout mechanisms, and candidate schedules.

Contents

1 Introduction 2

1.1 Optimal allocation of observation times 2

1.2 Design criteria and cost considerations 2

1.3 Practical guidance on the number of measures 3

1.4 Extensions and related work 3

1.5 Power and sample size tools 3

1.6 Present study 3

2 Methods 4

2.1 Data generating model 4

2.2 Covariance structure and theoretical prediction 4

2.3 Parameter values 5

2.4 Design specifications 6

2.5 Analytic relative efficiency 6

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36	2.6 Scenario extension (analytic only)	7
37	2.7 Simulation procedure	8
38	3 Results	8
39	3.1 Summary statistics	8
40	3.2 Distribution of estimates	9
41	3.3 Standard errors across simulations	9
42	4 Discussion	9
43	5 References	11

44 1 Introduction

45 In longitudinal clinical trials, subjects are measured repeatedly over a fixed study period.
46 The conventional approach assigns observation times at equally spaced intervals (monthly
47 or quarterly visits, for example) largely for administrative convenience rather than statistical
48 efficiency. A growing body of work in optimal experimental design suggests that the placement
49 of these observation times can substantially affect the precision with which treatment effects
50 are estimated. We begin in Section 1.1 with a review of the relevant literature and then
51 describe the analytic efficiency calculation and simulation study, and their results, in Sections
52 2 and 3.

53 1.1 Optimal allocation of observation times

54 ¹ provided the foundational analysis of this problem for two-group trials with linearly divergent
55 treatment effects. They showed that the optimal allocation of time-points depends critically
56 on the assumed covariance structure. Under compound symmetry (CS), efficiency improves
57 by clustering repeated measures near the end of the study period, where the treatment effect
58 is largest. Under a first-order autoregressive (AR(1)) structure, only two measurements –
59 at baseline and at study end – are needed for highly efficient estimation. Under a more
60 realistic AR(1) plus measurement error structure, the optimal spacing depends on the specific
61 covariance parameter values, but the equally spaced design is generally suboptimal.

62 In a companion paper,² extended these results to address the joint optimization of the number
63 of repeated measures and group sizes under budget constraints. They found that when bud-
64 get is fixed and the covariance structure includes an autoregressive component, two repeated
65 measures per subject often yield highly efficient treatment effect estimators. Otherwise, inter-
66 mediate measures can reduce the variance of the generalized least squares (GLS) estimator,
67 particularly under compound symmetry.

68 1.2 Design criteria and cost considerations

69 The formal framework for these results relies on classical optimal design theory, as developed
70 by^{3,4} proposed maximin D-optimal designs for longitudinal mixed effects models, computing
71 optimal allocations of time-points for polynomial trend models with random intercepts, ran-

dom slopes, and AR(1) serial correlations. Because D-optimal designs depend on unknown parameter values, the maximin criterion provides robustness by maximizing the worst-case relative efficiency across the parameter space.

⁵ extended this work to cohort designs, showing that a purely longitudinal design with a single cohort measured at optimal time-points is generally more efficient than mixed longitudinal-cross-sectional designs. Their numerical results demonstrated that the most efficient number of repeated measurements equals the sum of the number of cohorts and the degree of the polynomial trend. Subsequently,⁶ showed that adding cohorts or repeated measurements beyond this optimum wastes resources when the study budget is fixed.

1.3 Practical guidance on the number of measures

From a more pragmatic perspective,⁷ examined the marginal benefit of adding repeated measures in terms of statistical power. He showed that while repeating assessments can have dramatic effects on power, the marginal benefit of each additional measure decreases rapidly. In general, there is little value in exceeding four post-treatment assessments (or seven total including baseline), except when correlations between measures are low, as in episodic conditions.

1.4 Extensions and related work

Several extensions of the Winkens framework have appeared.⁸ generalized the optimal design results to second-order polynomial treatment effects, finding that equidistant time-points are either optimal or highly efficient when only three measures are taken.⁹ investigated optimal designs when the goal is to estimate individual growth curves (random effects) rather than fixed effects, showing that the optimal number of time-points and subjects differs from the fixed-effects case.¹⁰ developed Bayesian D-optimal designs for dichotomous repeated measures with autocorrelation, formally accounting for parameter uncertainty through prior distributions.

For the analysis of unequally spaced longitudinal data,¹¹ provided the statistical machinery for fitting mixed models with continuous-time AR(1) structures and arbitrary observation schedules. Their approach, based on the Kalman filter, remains foundational for analyzing data from designs with non-equidistant time-points.

1.5 Power and sample size tools

Modern software for power and sample size calculations in longitudinal trials has made these design considerations more accessible.¹² describe the **longpower** R package, which implements closed-form power calculations for linear mixed models.¹³ applied these methods to Alzheimer's disease Phase II trial design, demonstrating the practical impact of design choices on the ability to detect disease-modifying treatment effects.

1.6 Present study

The theoretical optimality results reviewed above establish that the best design depends on the assumed covariance structure, but they do not by themselves quantify how much efficiency

is at stake for a specific, realistic trial configuration, nor do they address the finite-sample behavior of the REML-based inference that practitioners actually use. This report addresses that translational gap for one configuration: a 100-patient, two-arm, 24-month trial with a linearly divergent treatment effect, analyzed under a random-intercept, random-slope model. We compare two five-visit designs:

- **Equal spacing:** observations at months 0, 6, 12, 18, and 24.
- **Clustered spacing:** observations clustered near the beginning and end of the study period (months 0, 2, 4, 20, 24). Under a random-slope covariance structure (Section 2.1), boundary-loaded time-points are efficient for estimating the slope contrast for a classical leverage reason – observations far from the mean time carry more information about a slope – which is distinct from the compound-symmetry mechanism that motivates clustering in¹; see Section 2.2 for the derivation.

We first derive the exact asymptotic (GLS) relative efficiency of the two designs in closed form, then use a common-random-number Monte Carlo study to assess whether finite-sample REML estimation and inference (bias, empirical versus model-based standard error, power, and 95% confidence interval coverage) reproduce the analytic prediction. We do not claim that either design is optimal: the theoretically optimal design implied by the cited literature is not computed here, and only two candidate schedules are compared. `R/simulation.R` in the accompanying compendium implements the data-generating model, the analytic efficiency calculation, and the paired simulation as reusable functions so the comparison can be extended to other covariance structures, sample sizes, dropout mechanisms, and candidate schedules (Section 2.6).

2 Methods

2.1 Data generating model

We simulate data from a linear mixed effects model with random intercepts and random slopes. Let Y_{ij} denote the outcome for subject i at time t_j . The generating model is:

$$Y_{ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})t_j + \beta_2 G_i + \beta_3 G_i t_j + \varepsilon_{ij}$$

where $G_i \in \{0, 1\}$ is the treatment group indicator, b_{0i} and b_{1i} are subject-specific random effects with $\text{Var}(b_0) = \tau_0^2$, $\text{Var}(b_1) = \tau_1^2$, $\text{Cov}(b_0, b_1) = \tau_{01}$, and $\varepsilon_{ij} \sim N(0, \sigma^2)$.

The parameter of interest is β_3 , the treatment-by-time interaction, which represents the rate at which the treatment effect diverges from the control effect per unit time.

2.2 Covariance structure and theoretical prediction

Marginalizing over the random effects, this model induces a within-subject covariance $\text{Cov}(Y_{ij}, Y_{ij'}) = \tau_0^2 + \tau_{01}(t_j + t_{j'}) + \tau_1^2 t_j t_{j'} + \sigma^2 \mathbb{1}(j = j')$, in which the marginal variance increases with t_j^2 and the correlation between visits depends on their product $t_j t_{j'}$, not on their lag $|t_j - t_{j'}|$. This is the standard random-coefficients (random-intercept, random-slope)

covariance structure. It is distinct from both compound symmetry (CS), which is the random-intercept-only special case ($\tau_1 = 0$) with constant correlation regardless of visit timing, and from a first-order autoregressive (AR(1)) structure, whose correlation depends only on the lag between visits and does not grow with τ_1 boundary leverage. An earlier draft of this report described the simulated structure as combining features of CS and AR(1); that characterization is incorrect and has been removed (see the Discussion for the corrected interpretation).

The efficiency prediction relevant to this structure differs from the CS-based argument in¹. Under CS, clustering observations near the end of the study helps because the treatment effect (and hence the signal) is largest there while every pair of visits is equally correlated, so nothing is lost by leaving a gap. Under the random-slope structure simulated here, the reason boundary-clustered designs are more efficient for the *slope* contrast β_3 is instead the classical leverage argument for regression slopes: for a simple linear regression of Y on t , $\text{Var}(\hat{\text{slope}}) \propto 1/\sum_i (t_i - \bar{t})^2$, so observations placed far from the mean time carry disproportionately more information about the slope. Section 2.5 below verifies this prediction is directionally correct but shows the efficiency gain shrinks as the random-slope variance τ_1^2 grows relative to the residual variance, because a larger τ_1^2 makes the slope itself (not just its estimation error) more variable across subjects, which increases the total variance contributed by every visit and dilutes the boundary-leverage advantage.

2.3 Parameter values

We use the following parameter values, broadly motivated by cognitive decline trials:

Table 1: Simulation parameters.

Parameter	Symbol	Value
Intercept	β_0	50.00
Time slope	β_1	-1.00
Group effect at baseline	β_2	0.00
Group x time	β_3	0.19
SD random intercept	τ_0	5.00
SD random slope	τ_1	0.30
Residual SD	σ	3.00
Subjects per group	n	50.00
Study duration (months)	–	24.00
Number of simulations	–	2000.00

The treatment-by-time interaction $\beta_3 = 0.19$ implies that the treatment group improves (or declines less) by 0.19 units per month relative to control, yielding a cumulative treatment advantage of 4.56 units over 24 months.

2.3.1 Recalibration of the treatment effect

An earlier draft used $\beta_3 = 0.5$, for which the analytic power of both designs is 1 to machine precision (Section 2.5): every replication rejects the null, so power cannot discriminate be-

172 tween designs and the metric is uninformative. We recalibrated β_3 so that the equal-spacing
 173 design has approximately 80% analytic power at the two-sided $\alpha = 0.05$ level, which requires
 174 $\beta_3/\text{SE}_{\text{equal}} \approx 2.80$ (the sum of the z -quantiles for 80% power and a two-sided 5% test). With
 175 $\text{SE}_{\text{equal}} = 0.0678$ (Section 2.5), this gives $\beta_3 \approx 0.19$, for which the analytic power is 80.0%
 176 (equal spacing) and 82.4% (clustered spacing) – close to, and not at, the ceiling, so that power
 177 remains informative while both designs are still adequately powered for a realistic trial. All
 178 other parameters are unchanged from the original design.

179 2.4 Design specifications

Table 2: Observation times (months) for each design.

Design	Visit 1	Visit 2	Visit 3	Visit 4	Visit 5
Equal	0	6	12	18	24
Clustered	0	2	4	20	24

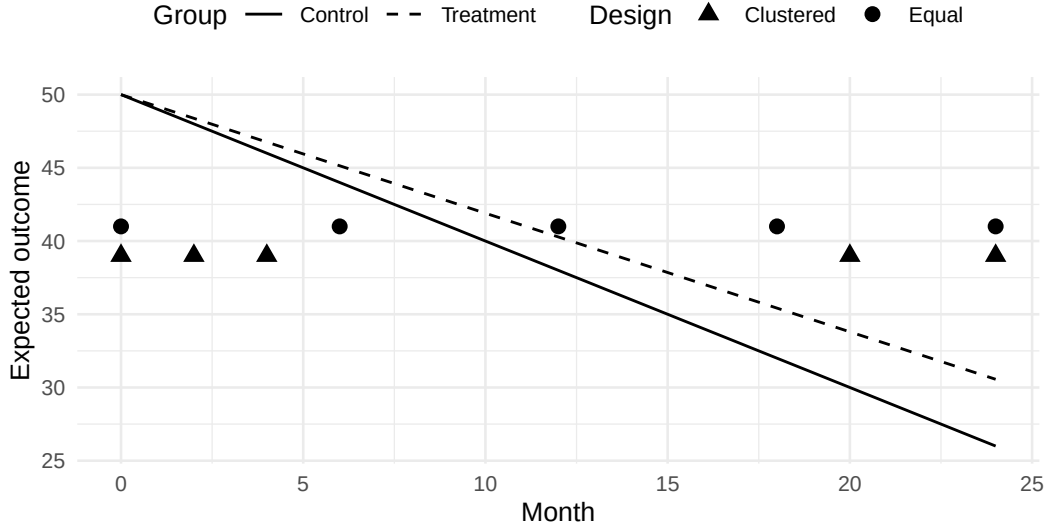


Figure 1: Schematic of the two visit schedules over the 24-month study period. Lines show the expected group trajectories; markers show each design’s visit times, vertically offset (Equal above, Clustered below) purely to avoid overplotting at shared visit times (0 and 24 months) and carry no other meaning.

180 2.5 Analytic relative efficiency

181 Under the data generating model of Section 2.1, the treatment-by-time interaction $\hat{\beta}_3$ has an
 182 exact large-sample GLS variance: with $\mathbf{Z}_i = [\mathbf{1}, \mathbf{t}]$ the $k \times 2$ per-subject design matrix for
 183 the random effects, $\mathbf{D}$ the random-effects covariance matrix, and $\mathbf{V} = \mathbf{Z}_i \mathbf{D} \mathbf{Z}_i' + \sigma^2 \mathbf{I}_k$, the
 184 information matrix $\mathbf{M} = \sum_i \mathbf{X}_i' \mathbf{V}^{-1} \mathbf{X}_i$ (summed separately over control subjects, for whom
 185 the last two columns of $\mathbf{X}_i$ are zero, and treatment subjects) gives $\text{Var}(\hat{\beta}_3) = [\mathbf{M}^{-1}]_{44}$. This
 186 ten-line linear algebra calculation, implemented as `analytic_gls_se()` in `R/simulation.R`,
 187 requires no simulation:

Table 3: Exact GLS standard errors of the treatment-by-time interaction under the random-slope data-generating model.

Design	Analytic SE
Equal	0.0678
Clustered	0.0658

The exact GLS standard errors are 0.0678 (equal spacing) and 0.0658 (clustered spacing), giving a relative efficiency (ratio of variances) of 1.063 in favor of the clustered design. This analytic value is the primary quantity of interest for the design comparison; the simulation in Section 2.7 exists to answer a different question – whether finite-sample REML estimation and Wald-type inference (bias, coverage, and empirical versus model-based SE) reproduce this analytic prediction and behave as expected at $n = 100$ – not to re-derive a number already available in closed form.

2.6 Scenario extension (analytic only)

The single scenario simulated in this report fixes one covariance structure, one effect size, and one sample size (2026-08-16 review, issue 2.4). Because the design comparison’s quantity of interest, the GLS relative efficiency, has a closed form (Section 2.5), `analytic_gls_se_general()` in `R/covariance.R` generalizes the calculation to an arbitrary within-subject covariance matrix, so the sensitivity of the efficiency contrast to the covariance structure can be assessed without additional Monte Carlo simulation. Table 4 reports the exact relative efficiency of the clustered versus equal-spacing design across the random-slope SD τ_1 used in the main simulation, and for two additional structures: compound symmetry, the mechanism that motivates clustering in¹, and AR(1) plus measurement error.

Table 4: Analytic (closed-form, no simulation) relative efficiency of the clustered versus equal-spacing design across covariance structures and random-slope variance.

Structure	Parameter	SE (Equal)	SE (Clustered)	Relative efficiency
Random slope	$\tau_1 = 0.10$	0.0374	0.0336	1.24
Random slope	$\tau_1 = 0.20$	0.0510	0.0482	1.12
Random slope	$\tau_1 = 0.30$	0.0678	0.0658	1.06
Random slope	$\tau_1 = 0.40$	0.0860	0.0844	1.04
Random slope	$\tau_1 = 0.50$	0.1049	0.1036	1.03
Compound symmetry	$\tau_0 = 5.0$	0.0316	0.0269	1.38
AR(1) + error	$\rho = 0.8$	0.0554	0.0528	1.10

The clustered design is more efficient across the full random-slope grid, but the advantage shrinks as τ_1 grows relative to the residual SD σ , consistent with the leverage-dilution argument of Section 2.2. Under compound symmetry and under AR(1) plus measurement error the clustered design is also more efficient, although by a different mechanism in each case. This extension is analytic only: it does not assess whether finite-sample REML estimation and inference reproduce these predictions for structures other than the random-slope structure actually simulated, nor does it vary sample size or dropout. Extending the Monte Carlo comparison itself to this grid, and to dropout, remains future work (see the Discussion); **Rscript**

213 `analysis/scripts/run_scenario_grid.R` regenerates Table 4.

214 2.7 Simulation procedure

215 The simulation functions (`simulate_trial()`, `run_one_sim()`, `run_paired_simulation()`,
216 `analytic_gls_se()`) live in R/ as documented, unit-tested package functions (`inst/tinytest/`)
217 rather than inline in this document, so they can be reused for the scenario extension and
218 audited independently of the manuscript (2026-08-16 review, issue 2.8). For each of 2000
219 Monte Carlo replications, `run_paired_simulation()`:

- 220 1. Generates 100 subjects, randomized 1:1 to treatment and control, drawing one shared
221 set of subject-level random effects (b_{0i}, b_{1i}) and one shared set of residual errors ε_{ij} per
222 subject-visit index $j = 1, \dots, 5$.
- 223 2. Builds the equal-spacing and clustered-spacing datasets from *those same draws*, so the
224 j -th random effect and residual realization is reused across designs for the j -th visit
225 rather than redrawn – these are common random numbers (CRN), which pair the two
226 designs within each replication and substantially reduce the Monte Carlo error of the
227 *difference* between designs relative to two independent simulation arms.
- 228 3. Fits a linear mixed model (random intercepts and slopes) to each design’s dataset using
229 restricted maximum likelihood (REML) via `nlme::lme()`.
- 230 4. Extracts the point estimate, standard error, and p -value for $\hat{\beta}_3$ for each design.

231 For each design we compute the mean estimate (bias), the empirical SD of $\hat{\beta}_3$, the mean
232 model-based SE, empirical power ($p < 0.05$), and 95% CI coverage, each with a Monte Carlo
233 standard error (MCSE) computed following¹⁴: MCSE of the empirical SD is $SE/\sqrt{2(R-1)}$,
234 MCSE of a proportion (power, coverage) is $\sqrt{\hat{p}(1-\hat{p})/R}$, and MCSE of the bias is $SE/\sqrt{R}$,
235 where R is the number of converged replications. We additionally report a paired bootstrap
236 (resampling replication indices, which respects the CRN pairing) 95% interval for the relative
237 efficiency (ratio of empirical variances).

238 The driver script `analysis/scripts/run_simulation.R` runs this procedure and saves the
239 raw per-replication estimates, the summary statistics with MCSEs, the analytic SEs, and the
240 bootstrap interval to a single versioned artifact, `analysis/data/derived_data/sim_results.rds`,
241 which this document reads directly rather than re-simulating on every render or relying on
242 the knitr cache to stay synchronized with the reported numbers (2026-08-16 review, issue
243 2.8(e)). To regenerate: `Rscript analysis/scripts/run_simulation.R 2000`. The results
244 below used 2000 replications, R version 4.6.1 (2026-06-24), and `nlme` 3.1.169, completing in
245 176 seconds (single core, no parallelization).

246 3 Results

247 3.1 Summary statistics

248 Replications for which `nlme::lme()` failed to converge (an error or warning during fitting,
249 per `run_one_sim()` in `R/simulation.R`) are recorded as NA and excluded from the summary

Table 5: Simulation results for the treatment-by-time interaction, with Monte Carlo standard errors (MCSE) of each metric computed following Morris, White, and Crowther (2019).

Design	Converged	Mean Est.	Bias	MCSE(Bias)	Emp. SE	MCSE(Emp. SE)	Model SE	Power	MCSE(Power)	Coverage	MCSE(Coverage)
Equal	2000	0.189	-0.0011	0.0015	0.0669	0.0011	0.0676	0.787	0.0092	0.952	0.0048
Clustered	2000	0.189	-0.0015	0.0014	0.0646	0.0010	0.0655	0.814	0.0087	0.957	0.0046

statistics above rather than imputed or retried; the **Converged** column reports how many of the 2000 attempted replications per design contributed to each summary statistic.

We observe that the equal-spacing design achieves an empirical power of 78.7% (MCSE 0.9%), compared with 81.3% (MCSE 0.9%) for the clustered design; both are informative rather than at ceiling following the recalibration of $\hat{\beta}_3$ (Section 2.3.1). The empirical standard error of $\hat{\beta}_3$ is 0.0669 (MCSE 0.00106) under equal spacing and 0.0646 (MCSE 0.00102) under the clustered design. The paired-bootstrap relative efficiency (ratio of empirical variances, resampling replication indices so the common-random-number pairing is respected) is 1.07, 95% bootstrap interval 1.06-1.09, consistent with the analytic relative efficiency of 1.063 (Section 2.5); the interval excludes 1, so the precision advantage of the clustered design is distinguishable from Monte Carlo noise.

3.2 Distribution of estimates

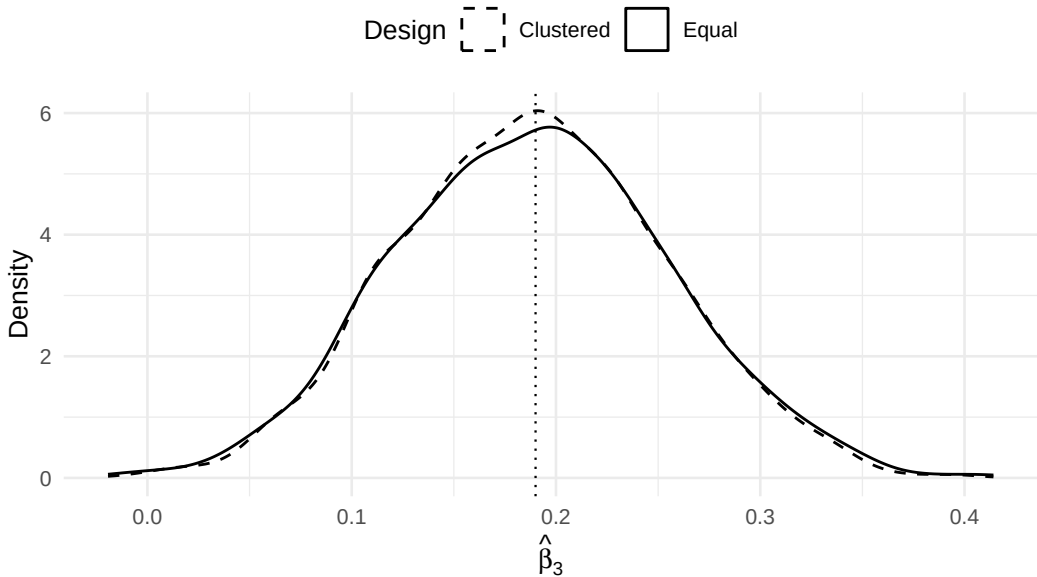


Figure 2: Sampling distribution of the treatment-by-time interaction estimate under each design.

3.3 Standard errors across simulations

4 Discussion

In this study, we have endeavored to illustrate the practical consequences of visit placement in longitudinal clinical trials. Both designs use the same number of subjects and the same

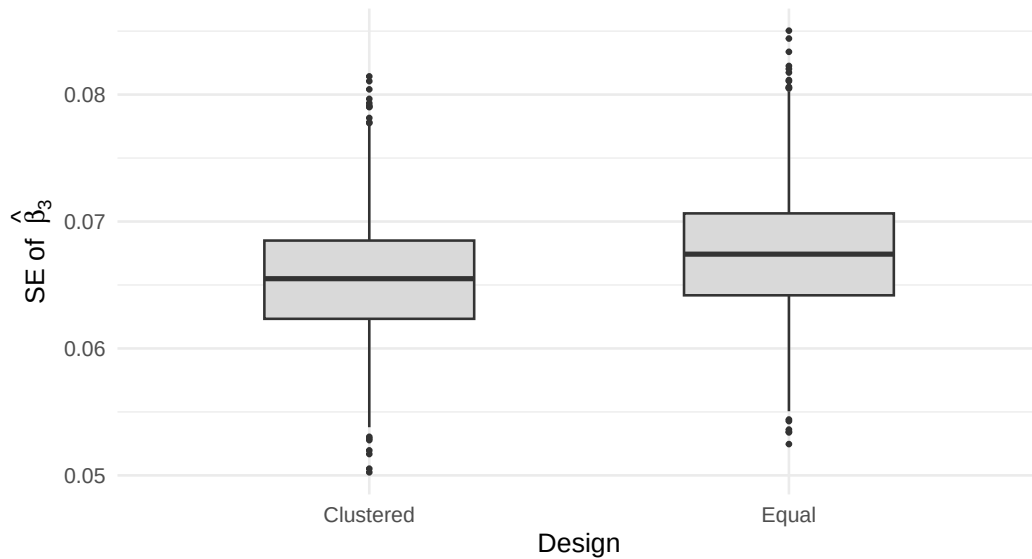


Figure 3: Distribution of model-based standard errors across simulations.

number of visits; they differ only in the timing of observations.

The results are broadly consistent, in direction, with the theoretical findings of¹, but not through the mechanism that paper’s compound-symmetry analysis invokes. As derived in Section 2.2, the data-generating mechanism used here is the standard random-intercept, random-slope covariance structure, not a structure that “combines features of compound symmetry and AR(1) with measurement error” as an earlier draft of this report incorrectly stated; it contains no autoregressive component. The clustered design’s precision advantage for the slope contrast β_3 follows instead from the classical leverage argument for regression slopes. The advantage is not an artifact of Monte Carlo noise: the paired bootstrap relative efficiency (ratio of empirical variances) is 1.07 (95% bootstrap interval 1.06 to 1.09), consistent with the analytic relative efficiency of 1.063 from Section 2.5, and the interval excludes 1.

Several caveats warrant mention. First, we assumed complete data with no dropout. In practice, attrition is common in longitudinal trials and may differentially affect designs that place observations at extreme time-points. Second, the data-generating model used independent random effects and homoscedastic residuals; more complex covariance structures (e.g., heterogeneous variances, serial correlation in residuals) could alter the relative performance of the two designs. Third, we considered only a linear treatment effect;⁸ showed that optimal designs differ for polynomial effects of higher order.

In conclusion, three points merit emphasis. First, the placement of observation times in a longitudinal trial is a design decision with real consequences for statistical efficiency, and the default of equal spacing is not in general optimal. Second, the form of the covariance structure is critical: results that hold under compound symmetry may not carry over to random-slope models or AR(1) structures. Third, the robustness of these findings to dropout, covariance misspecification, and nonlinear treatment trajectories remains to be investigated. The maximin D-optimal framework of⁴ and the Bayesian approach of¹⁰ provide principled

291 starting points for that work.

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